KRISHNA JAIN

ACADEMIC DETAILS

Year	Degree / Board	Institute	GPA / Marks(%)
---	B.Tech in Mechanical Engineering	Indian Institute of Technology Delhi	7.95
2024	CBSE	MB International School ,Kota	94.2
2022	CBSE	White Leaf Public School,Delhi	93.8

SCHOLASTIC ACHIEVEMENTS

- **INAO 2023 Qualifier** – Qualified for the Indian National Astronomy Olympiad (Class XI).
- **TALLENTEX 2022** – Secured **Rank 10** and was awarded allen scholarship.(Class X)
- **MVPP Scholarship Recipient (2020–21)** – Awarded the Mukhya Mantri Vigyan Pratibha Pariksha Scholarship (formerly JSTSE) by the Government of Delhi (Class IX).

PROJECTS

- **Rubik's Cube Solver based on a custom two-phase algorithm** (Self) *(2026-Ongoing)*
  -Developed an edge-first, corner-second solving strategy with a guarantee of solving every reachable cube configuration.
  -Implemented heuristic search using IDA* with pruning strategies for optimal cube solving.
  -Successfully solved 100,000 randomly generated cube configurations with 100% correctness.
  - Evaluated the custom solver against Thistlethwaite-based solvers using move count, execution time, and scalability.
- **GenAI Agentic Systems**(CSoT-2026) *(June 2026-Current)*
  -Built an AI assistant integrating web search, document retrieval, Python execution, and external tools for complex tasks.
  -Designed a modular pipeline integrating 6 components, enabling independent development, testing, and replacement of each module.
- **Machine Learning Assignments** *(2026-Ongoing)*
  1.**Background Segmentation and Eigenspaces**-
  -Implemented PCA-based video summarization which reduced video length by 80% while preserving key events.
  -Implemented GMM background subtraction algorithm to extract key frames and separate foreground from background using OpenCV.
  2.**Digit Classification**-
  -Achieved 98.2% MNIST accuracy using a fully connected neural network after systematic hyperparameter optimization.
  -Built a Transformer-based digit classifier achieving 99.1% validation accuracy, outperforming the MLP baseline.
  -Compared neural network architectures through accuracy, training time, and parameter efficiency across multiple experiments.
- **Edmonds' Blossom Algorithm for Maximum Matching** *(Spring 2026)*
  - Implemented Edmonds' Blossom algorithm to compute maximum matchings in general graphs using blossom contraction.
  - Visualized augmenting paths and blossom contractions, validating correctness across diverse non-bipartite graph test cases.
- **Autonomous Aerial Systems** (CSoT-2026) *(June 2026-Aug 2026)*
  -Designed a quadcopter using a model-based design workflow, 6-DOF dynamics, sensor models and PID flight control.
  -Developed a state estimator using an Extended Kalman Filter for IMU and GPS fusion.
- **Chinese Postman Solver for Road Networks** *(Spring 2026)*
  - Implemented a Chinese Postman solver using shortest paths, Eulerian augmentation, and graph matching techniques.
  - Generated minimum-cost traversal routes for OpenStreetMap road networks through efficient graph preprocessing.
- **Quantitative Finance** (CSoT-2026) *(June 2026-Current)*
  -Implemented mean-variance portfolios achieving 19.4% annualized return, 1.27 Sharpe ratio, and 13% maximum drawdown.
  -Reduced backtesting bias by eliminating look-ahead and survivorship bias during strategy evaluation.
- **Human Hand & Wrist Prototype**(Self) *(Autumn 2025)*
  - -Modeled a biomimetic human hand and wrist in SolidWorks with articulated joints, motion simulations, and structural analysis.
  - -Stimulated different alphabets and shapes through finger and hand motion.
- **Portable Vacuum Cleaner** (Prof. Sagar Sarkar) *(Spring 2025)*
  -Designed and fabricated an ESP32-based vacuum cleaner with ultrasonic sensing.

TECHNICAL SKILLS

- **Programming Languages**: C++, Python, MATLAB
- **Libraries & Technologies**: NumPy, Pandas, , OpenAI API, MCP.
- **Simulation and CAD**-Solidworks,Ansys Fluent,Ansys Mechanical and MATLAB/Simulink.

EXTRA CURRICULAR ACTIVITIES

- Secured **Runner-Up** position in the Ansys PyFluentHackathon conducted in February 2026.
- Participated in **NCC winter camp 2024**
- Got **Grade A** in Intermediate Drawing Grade Exam
- Passed with **First Class** in Tabla Certification Exam

Disclaimer: All the information on this page is entered by student.

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IIT COURSE

Degree	Institute	CGPA	Dept. Rank
B.Tech in Mechanical Engineering	Indian Institute of Technology Delhi	7.95	22

COURSES DONE

Language & Writing Skill, Intro. To Computer Science, Physics Laboratory, Electromagnetic Waves & Qua.mec., Intro. To Electrical Engg., Intro. To Electrical Engg., Calculus, Fluid Mechanics, Engineering Thermodynamics, Kinematics & Dy. Of Machines, Solid Mechanics, Introduction To Psychology, Introduction To Materials Science And Engineering, Linear Algebra & Diffe. Equa., Engineering Mechanics, Product Realization Through Manufacturing, Chemistry Laboratory, Introduction To Chemistry, Energy Systems And Tech., Introduction To Statistics, Data Structures And Algorithms, Mechanical Engg. Drawing

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